

Conductor Descent and Flat Distribution Ranks

Stieltjes jets, cyclotomic polylogarithms,
and prime-support vanishing

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Abstract

The Polylogarithms manuscript reports a finite distribution-rank pattern for parameter derivatives of generalized Stieltjes constants, but does not give a uniform proof. We define the complete finite-level distribution presentation and prove a stronger statement: over a characteristic-zero splitting field, its multiparameter deformation is a free module of rank $\varphi(q)$, with one explicit generator for every primitive character whose conductor divides q . Consequently the endpoint-fixed rank is $q - \varphi(q)$ for every denominator, every specialization of the prime weights, and every spectral order. The result persists for arbitrary jets, with no resonance-induced rank jump.

The same conductor coordinates give finite Möbius formulas for every Stieltjes parameter-derivative layer and for character-weighted cyclotomic polylogarithm traces. In particular, a principal primitive-root trace has an exact zero of order $\omega(q) - 1$ at polylogarithm order one. This produces explicit logarithmic identities for order derivatives, including a second order identity at level 30 and a twisted identity at level 260. We also give a small exact certificate showing that the manuscript's S_4 candidate is not a consequence of a specified weight-five single-product double-shuffle presentation. The candidate itself is not proved or disproved here. All dimension statements concern presented formal quotients, not numerical independence of periods.

Status. The theorems below have ordinary mathematical proofs. Finite matrix and polynomial statements have replayable exact-arithmetic checks. Floating-point tests are supplementary, not interval certificates. This note has not been independently refereed or formalized in a proof assistant. The rank phenomenon is connected to classical universal-distribution theory; no claim of global priority is made for it or for identities that follow from standard character and Euler-factor formulas.

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1 The manuscript target and the scope of the contribution

The source is the consolidated manuscript *Polylogarithms and their Arithmetic Bridges* in the ProveIt repository [1]. Its Chapter 8, Observation `tower:thm:rank`, records an exact finite experiment for $3 \leq q \leq 30$ and parameter-derivative orders $k = 1, 2, 3$:

$$(q - 1) - \text{rank} = \varphi(q) - 1. \quad (1.1)$$

The source explicitly states that a uniform proof is absent and that the historical scripts were not imported. Chapter 10 lists the divisor-row structure as a remaining proof problem.

The present work addresses that gap with a fully specified matrix. We do not identify an unseen historical matrix with our matrix by assumption. Rather, we prove the rank statement for the natural complete distribution system, provide its generator and tests, and supply replacement manuscript text that states the presentation explicitly. Reconciliation with the historical scripts, should they be recovered, is a separate editorial task.

The decisive observation is that primitive residues at denominator q are not the most robust coordinates. The right coordinates are character sums at their *minimal conductors*. In those coordinates, every additional prime produces a polynomial Euler factor. No division by that factor is necessary. Thus zeros of Euler factors can destroy a particular coordinate chart without changing the rank of the distribution module.

Our main outputs are as follows. Theorem 3.1 proves a multiparameter free-module normal form. Corollary 3.3 and Theorem 4.1 give the uniform rank and jet counts. Theorem 5.1 gives explicit all-order Stieltjes descent. Theorems 6.1 and 7.1 give the cyclotomic trace identities and their exact vanishing orders. Proposition 10.2 records a distinct finite-presentation obstruction relevant to the unproved S_4 candidate.

1.1 Relation to existing mathematics

Kubert proved freeness and primitive-cardinality rank for the universal ordinary distribution, including its rational representation structure [2]. Ouyang developed a broader universal norm-distribution framework [3]. Our polynomial deformation belongs to this mathematical setting. The purpose here is an elementary, explicit conductor-coordinate derivation tailored to the manuscript's rows, together with jet formulas and machine-replayable certificates. The general idea of using characters to diagonalize distributions is not new.

The underlying Hurwitz multiplication law, Laurent expansion and root-of-unity Fourier formulas are standard [5, 6]. The parameter-derivative ladder is already present in the manuscript and in Coffey's all-orders work [4]; it is not claimed as a new result of this note. The polylogarithm identities below are derived explicitly, with special attention to imprimitive characters and the removable singularity at order one.

1.2 Two kinds of derivatives

We use $\zeta^{(n)}(s, a) = \partial_s^n \zeta(s, a)$ for spectral derivatives. The notation $\gamma_n^{(k)}(a) = \partial_a^k \gamma_n(a)$ denotes a parameter derivative. To avoid confusion, put

$$\mathcal{L}^{[j]}(z) := \left. \frac{\partial^j}{\partial s^j} \text{Li}_s(z) \right|_{s=1}. \quad (1.2)$$

Thus $\mathcal{L}^{[1]}(z)$ is an order derivative, not $z d\text{Li}_1(z)/dz$ and not $\text{Li}_0(z)$. This distinction is essential in the new explicit examples.

2 A precise finite-level distribution presentation

Fix $q \geq 2$. Let $\mathcal{U}_d = (\mathbb{Z}/d\mathbb{Z})^\times$ for $d > 1$, represented by $1 \leq a \leq d$ with $(a, d) = 1$, and put $\mathcal{U}_1 = \{1\}$. The fraction $1/1$ represents the endpoint, or residue zero modulo one.

Let K be a characteristic-zero field containing the values of all Dirichlet characters modulo q . Introduce an independent indeterminate t_p for each prime $p \mid q$ and set

$$R_q = K[t_p : p \mid q].$$

Start with the free R_q -module on symbols $E(a/d)$, where $d \mid q$ and $a \in \mathcal{U}_d$. There are exactly q symbols because $\sum_{d \mid q} \varphi(d) = q$. Fractions are always reduced before symbols are compared.

For $pd \mid q$, p prime and $a \in \mathcal{U}_d$, impose

$$\sum_{j=0}^{p-1} E\left(\frac{a+jd}{pd}\right) - t_p E(a/d) = 0. \quad (2.1)$$

Call the quotient $\mathcal{D}_q$. These are formal relations. Substituting $t_p = p^s$ realizes them by Hurwitz zeta values through

$$\sum_{j=0}^{p-1} \zeta\left(s, \frac{x+j}{p}\right) = p^s \zeta(s, x),$$

initially for $\operatorname{Re} s > 1$ and then meromorphically.

Definition 2.1 (The matrix used in the rank theorem). Index the q columns by $E(a/q)$, $1 \leq a \leq q$, with fractions reduced when necessary. For every prime $p \mid q$ and $1 \leq b \leq q/p$, the row is

$$\sum_{j=0}^{p-1} e_{b+jq/p} - t_p e_{pb}. \quad (2.2)$$

The resulting matrix is $M_q(\mathbf{t})$. The endpoint-fixed matrix $\widehat{M}_q(\mathbf{t})$ is obtained by deleting column q , not by deleting rows that involve the endpoint.

Lemma 2.2 (Prime rows suffice). *If $t_d = \prod_{p \mid d} t_p^{v_p(d)}$, every composite distribution row*

$$\sum_{j=0}^{d-1} E((x+j)/d) - t_d E(x)$$

whose arguments have denominator dividing q is generated by the prime rows. Conversely, the complete all-divisor presentation includes all prime rows.

Proof. For $d = uv$, first distribute each of the u preimages through a v -fold cover, then sum. The uv preimages occur once each, while the scalar coefficient becomes $t_u t_v = t_{uv}$. Subtracting the successive relations expresses the composite row as a row combination. Induct on the number of prime factors counted with multiplicity. All intermediate denominators divide the final denominator, so they remain within the finite level. $\square$

The row with parent argument 1 is indispensable. For a prime level $q = \ell$, it is the only row and reads

$$\sum_{a=1}^{\ell-1} E(a/\ell) + (1 - t_\ell) E(1) = 0. \quad (2.3)$$

After fixing $E(1)$, the rank is one. Omitting this row would change the residual dimension by one and would not model the stated experiment.

3 A conductor-adapted free basis

Every character modulo q is induced by a unique primitive character χ^* of conductor $f \mid q$. For $f \mid d \mid q$, write

$$Z_{d,\chi^*} = \sum_{a \in \mathcal{U}_d} \chi^*(a) E(a/d). \quad (3.1)$$

At $f = 1$ the primitive character is the identically-one character, and $Z_{1,1} = E(1)$. At other f , the primitive character is extended by zero on integers not coprime to f .

Define the polynomial

$$A_{d,f,\chi^*}(\mathbf{t}) = \prod_{p \mid f} t_p^{v_p(d) - v_p(f)} \prod_{\substack{p \mid d \\ p \nmid f}} t_p^{v_p(d) - 1} (t_p - \chi^*(p)). \quad (3.2)$$

An empty product is one.

Theorem 3.1 (Polynomial conductor normal form). *The module $\mathcal{D}_q$ is free over R_q of rank $\varphi(q)$. A basis is*

$$\{Z_{f,\chi^*} : f \mid q, \chi^* \text{ primitive of conductor } f\}. \quad (3.3)$$

For every $f \mid d \mid q$,

$$Z_{d,\chi^*} = A_{d,f,\chi^*}(\mathbf{t}) Z_{f,\chi^*}. \quad (3.4)$$

In particular, this normal form remains valid after arbitrary specialization of the prime weights, including zero weights and roots of unity.

Proof. The group $G = \mathcal{U}_q$ acts by multiplying numerators. Since K contains all character values and $|G|$ is invertible, the usual character idempotents split the free module into character components. A denominator- d component contains the character induced by χ^* exactly when $f \mid d$, and then its corresponding eigenspace has rank one, generated by Z_{d,χ^*} . Equivalently, finite character orthogonality gives

$$E(a/d) = \frac{1}{\varphi(d)} \sum_{\chi \bmod d} \overline{\chi(a)} Z_{d,\chi^*}. \quad (3.5)$$

Here each χ is replaced on units by its inducing primitive character.

We compute the prime relation in one character component. Sum (2.1) over $a \in \mathcal{U}_d$ with weight $\chi^*(a)$. If $p \mid d$, every numerator $a + jd$ is coprime to pd , and every unit modulo pd appears once. Therefore

$$Z_{pd,\chi^*} = t_p Z_{d,\chi^*} \quad (p \mid d). \quad (3.6)$$

If $p \nmid d$, exactly one of the p numerators is divisible by p . After reducing that fraction, its contribution is

$$\sum_{b \in \mathcal{U}_d} \chi^*(pb) E(b/d) = \chi^*(p) Z_{d,\chi^*}.$$

The other numerators are precisely the units modulo pd . Hence

$$Z_{pd,\chi^*} = (t_p - \chi^*(p)) Z_{d,\chi^*} \quad (p \nmid d). \quad (3.7)$$

Every original row is a linear combination of these character-summed rows, by Fourier inversion on $\mathcal{U}_d$. No character with $f \nmid d$ occurs in that row module, so no additional relation is missed.

Now fix χ^* . Its available denominators form the divisor interval $f \mid d \mid q$. Relations (3.6)–(3.7) express every Z_{d,χ^*} as a multiple of Z_{f,χ^*} by moving upward from f . The multiplier for a new

prime is $t_p - \chi^*(p)$; every further copy of the same prime contributes t_p . Multipliers from distinct primes commute, so the result is independent of the chosen path and is exactly (3.2).

To prove freeness rather than merely spanning, map this character component to R_q by $Z_{d,\chi^*} \mapsto A_{d,f,\chi^*}$. Every edge relation maps to zero, and the proposed generator maps to one. Thus the component is isomorphic to R_q , with no additional relation on its generator. Taking the direct sum over characters of G proves freeness of rank $|G| = \varphi(q)$ and the claimed basis. The construction is polynomial, so specialization requires no nonvanishing assumption. $\square$

Remark 3.2 (What the conductor adds to a character transform). A character transform only at level q yields Z_{q,χ^*} , which can vanish formally when $A_{q,f,\chi^*} = 0$. The generator of the free component is instead Z_{f,χ^*} . Forgetting this distinction creates an apparent rank jump at a singular Euler factor. The normal form proves that the jump belongs to the chosen coordinates, not to the distribution quotient.

Corollary 3.3 (Uniform distribution rank). *For every specialization $t_p \in \mathbb{C}$,*

$$\text{rank } M_q(\mathbf{t}) = q - \varphi(q), \quad \text{rank } \widehat{M}_q(\mathbf{t}) = q - \varphi(q). \quad (3.8)$$

Thus the endpoint-fixed quotient has dimension $\varphi(q) - 1$. For $t_p = p^{k+1}$, the ranks are ranks over $\mathbb{Q}$ and hold for every integer $k \geq 1$ and every $q \geq 2$.

Proof. The specialized free quotient has dimension $\varphi(q)$, proving the first rank formula. Its endpoint $E(1) = Z_{1,1}$ is one of the conductor basis vectors. Thus quotienting further by $E(1)$ removes exactly one dimension. Equivalently, projection deleting the endpoint column is injective on the row space: a row-space element supported only on the endpoint would give a nonzero relation on this basis vector. This proves the second formula. For integer k , all entries are rational, and extending the coefficient field to $\mathbb{C}$ does not change matrix rank. $\square$

3.1 A fully explicit polynomial example at level twelve

Let χ_3 be the nontrivial character modulo 3, χ_4 the nontrivial character modulo 4, and $\chi_{12} = \chi_3\chi_4$. The four conductor generators are

$$Z_1 = E(1), \quad Z_3 = E(1/3) - E(2/3), \quad (3.9)$$

$$Z_4 = E(1/4) - E(3/4), \quad Z_{12} = E(1/12) - E(5/12) - E(7/12) + E(11/12). \quad (3.10)$$

For example,

$$\begin{aligned} E(1/12) = \frac{1}{4} [& t_2(t_2 - 1)(t_3 - 1)Z_1 + t_2(t_2 + 1)Z_3 \\ & + (t_3 + 1)Z_4 + Z_{12}], \end{aligned} \quad (3.11)$$

$$E(1/6) = \frac{1}{2} [(t_2 - 1)(t_3 - 1)Z_1 + (t_2 + 1)Z_3], \quad (3.12)$$

$$E(1/4) = \frac{1}{2} [t_2(t_2 - 1)Z_1 + Z_4]. \quad (3.13)$$

The remaining rows follow by character signs and denominator reduction. The artifact `q12_polynomial_normal_form.json` contains the entire 12×4 matrix F and the 4×12 extraction matrix B . The exact checks are

$$M_{12}F = 0, \quad BF = I_4. \quad (3.14)$$

They hold as polynomial identities in independent t_2, t_3 , not merely at numerical weights.

4 Jets and the uniform Stieltjes rank theorem

Let $s_0 \in \mathbb{C}$ and $N \geq 0$. For a family of symbols $X_a(s)$ satisfying Hurwitz-type distribution, introduce divided Taylor coefficients $X_{a,n} = [\varepsilon^n]X_a(s_0 + \varepsilon)$. The prime relations are

$$\sum_{j=0}^{p-1} X_{b+jq/p,n} - p^{s_0} \sum_{r=0}^n \frac{(\log p)^r}{r!} X_{pb,n-r} = 0, \quad 0 \leq n \leq N. \quad (4.1)$$

All powers use the real logarithm of the positive integer p .

Theorem 4.1 (No rank loss under arbitrary jets). *The complete system (4.1) on $(N+1)q$ coordinates has rank*

$$(N+1)(q - \varphi(q)). \quad (4.2)$$

Its solution space has dimension $(N+1)\varphi(q)$. Fixing all endpoint jets leaves $(N+1)(\varphi(q) - 1)$ free coordinates and does not change the row rank. These statements hold for every complex s_0 .

Proof. Work with truncated germs modulo ε^{N+1} and substitute $t_p = p^{s_0} \exp(\varepsilon \log p)$ into Theorem 3.1. Equivalently, choose arbitrary truncated germs for the conductor-character coordinates and define all other character germs by (3.4); then use Fourier inversion. This produces every solution of (4.1), because the character-summed equations still have unit coefficient on the higher-denominator coordinate. Extracting the minimal-conductor sums recovers the chosen germs, proving both completeness and uniqueness. There are $\varphi(q)$ freely chosen germs, each of length $N+1$. The endpoint is precisely the principal-conductor germ and can be fixed independently. Subtracting the solution-space dimension from the number of columns gives the rank. $\square$

The proof also applies when the quantities replacing $\log p$ are arbitrary complex numbers, or independent formal variables. In particular, no algebraic independence of logarithms is being used.

The generalized Stieltjes constants are defined by

$$\zeta(1 + \varepsilon, a) = \frac{1}{\varepsilon} + \sum_{n \geq 0} \frac{(-1)^n}{n!} \gamma_n(a) \varepsilon^n. \quad (4.3)$$

For $k \geq 1$, differentiating in a removes the pole and gives the known generating ladder

$$\sum_{n \geq 0} \frac{(-1)^n}{n!} \gamma_n^{(k)}(a) \varepsilon^n = (-1)^k (1 + \varepsilon)_k \zeta(k + 1 + \varepsilon, a). \quad (4.4)$$

For completeness, $\partial_a^k \zeta(s, a) = (-1)^k (s)_k \zeta(s + k, a)$ follows by termwise differentiation for $\operatorname{Re} s > 1$ and then continuation. Substitution into (4.3) proves (4.4).

Corollary 4.2 (The manuscript's rank pattern, uniformly). *Fix $q \geq 2$, $k \geq 1$ and $n \geq 0$. Treat endpoint values and lower Stieltjes indices as known. The complete distribution system on*

$$\gamma_n^{(k)}(a/q), \quad 1 \leq a < q,$$

has coefficient rank $q - \varphi(q)$ and formal residual dimension $\varphi(q) - 1$. If indices $0 \leq n \leq N$ are retained together, the residual dimension is $(N+1)(\varphi(q) - 1)$.

Proof. Multiplication by $(1 + \varepsilon)_k$ is invertible on truncated germs since its constant coefficient is $k! \neq 0$. It is common to every argument and therefore preserves distribution. The top-index coefficient matrix is $\widehat{M}_q$ with $t_p = p^{k+1}$. Lower-index logarithmic terms belong to the inhomogeneous right-hand side. Apply Corollary 3.3 and Theorem 4.1. $\square$

This proves a uniform statement about the explicit law presentation. It does not prove that the evaluated constants occupy that many independent numerical directions. Analytic functional equations or unrecorded period relations may reduce the evaluated span further.

5 Finite conductor descent for all Stieltjes layers

For a primitive character χ^* of conductor $f \mid d$, define

$$G_{d,\chi^*}^{(n,k)} = \sum_{a \in \mathcal{U}_d} \chi^*(a) \gamma_n^{(k)}(a/d). \quad (5.1)$$

Set

$$R(d, f) = \prod_{\substack{p \mid d \\ p \nmid f}} p, \quad m_r = \frac{d}{fr} \quad (r \mid R(d, f)). \quad (5.2)$$

Each m_r is a positive integer. Upon substituting $t_p = p^s$, (3.2) becomes the entire exponential polynomial

$$\begin{aligned} A_{d,f,\chi^*}(s) &= \left(\frac{d}{f}\right)^s \prod_{\substack{p \mid d \\ p \nmid f}} (1 - \chi^*(p)p^{-s}) \\ &= \sum_{r \mid R(d,f)} \mu(r) \chi^*(r) m_r^s. \end{aligned} \quad (5.3)$$

In particular,

$$A_{d,f,\chi^*}^{(j)}(s) = \sum_{r \mid R(d,f)} \mu(r) \chi^*(r) m_r^s \log^j m_r. \quad (5.4)$$

Theorem 5.1 (All-index, all-parameter-order descent). *For $n \geq 0$, $k \geq 1$ and $f \mid d$,*

$$\begin{aligned} G_{d,\chi^*}^{(n,k)} &= \sum_{r \mid R(d,f)} \mu(r) \chi^*(r) m_r^{k+1} \\ &\quad \cdot \sum_{j=0}^n (-1)^j \binom{n}{j} \log^j m_r G_{f,\chi^*}^{(n-j,k)}. \end{aligned} \quad (5.5)$$

No Euler-factor denominator occurs. Equivalently,

$$G_{d,\chi^*}^{(n,k)} = \sum_{j=0}^n (-1)^j \binom{n}{j} A_{d,f,\chi^*}^{(j)}(k+1) G_{f,\chi^*}^{(n-j,k)}. \quad (5.6)$$

Proof. For $\operatorname{Re} s > 1$, sorting a Dirichlet series by residue classes gives

$$\sum_{a \in \mathcal{U}_d} \chi^*(a) \zeta(s, a/d) = d^s L(s, \chi_d),$$

where χ_d is the induced character modulo d . Removing the extra Euler factors from its primitive L -series yields

$$L(s, \chi_d) = L(s, \chi^*) \prod_{\substack{p|d \\ p \nmid f}} (1 - \chi^*(p)p^{-s}).$$

Thus the character sum at d is $A_{d,f,\chi^*}(s)$ times the character sum at f . This is also the analytic realization of Theorem 3.1.

Insert $s = k + 1 + \varepsilon$, multiply by the common factor $(-1)^k(1 + \varepsilon)_k$, and use (4.4). Comparison of $(-1)^n \varepsilon^n / n!$ gives (5.6); the sign $(-1)^j$ comes from the Stieltjes Laurent convention. Finally apply (5.4). $\square$

5.1 Concrete second-index identities at level twelve

The multipliers are

$$A_{12,3,\chi_3}(s) = 4^s + 2^s, \quad A_{12,4,\chi_4}(s) = 3^s + 1.$$

Therefore, at $n = 2$, $k = 1$,

$$G_{12,\chi_3}^{(2,1)} = 20G_{3,\chi_3}^{(2,1)} - 72 \log 2 G_{3,\chi_3}^{(1,1)} + 68 \log^2 2 G_{3,\chi_3}^{(0,1)}, \quad (5.7)$$

$$G_{12,\chi_4}^{(2,1)} = 10G_{4,\chi_4}^{(2,1)} - 18 \log 3 G_{4,\chi_4}^{(1,1)} + 9 \log^2 3 G_{4,\chi_4}^{(0,1)}. \quad (5.8)$$

For instance,

$$G_{12,\chi_3}^{(2,1)} = \gamma'_2(1/12) - \gamma'_2(5/12) + \gamma'_2(7/12) - \gamma'_2(11/12),$$

and $G_{3,\chi_3}^{(2,1)} = \gamma'_2(1/3) - \gamma'_2(2/3)$. These are exact identities between explicitly specified signed sums; they require no integer-relation search. The two identities were also checked numerically using independent finite Hurwitz character sums.

5.2 The undifferentiated layer and its pole correction

For $k = 0$, a principal character sum still contains the Hurwitz pole. The same formula without a correction would be wrong.

Proposition 5.2 (The $k = 0$ correction). *For nonprincipal primitive χ^* the formula (5.6) remains valid with $k = 0$. For $f = 1$, put $A_d(s) = A_{d,1,1}(s)$. Then*

$$G_{d,1}^{(n,0)} = \sum_{j=0}^n (-1)^j \binom{n}{j} A_d^{(j)}(1) \gamma_{n-j}(1) + \frac{(-1)^n}{n+1} A_d^{(n+1)}(1). \quad (5.9)$$

Proof. For a nonprincipal character, the sum of its values over units is zero, so the $1/\varepsilon$ residues cancel. Coefficient comparison is unchanged. For the principal conductor, multiply

$$A_d(1 + \varepsilon) \left(\frac{1}{\varepsilon} + \sum_{n \geq 0} \frac{(-1)^n \gamma_n(1)}{n!} \varepsilon^n \right).$$

The pole term contributes $A_d^{(n+1)}(1)/(n+1)!$ to the coefficient of ε^n . Multiplying by $(-1)^n n!$ gives the correction in (5.9). The residue itself is $A_d(1) = d \prod_{p|d} (1 - p^{-1}) = \varphi(d)$, as required. $\square$

For the non-character multiplication formula this specializes to

$$\sum_{j=0}^{d-1} \gamma_n\left(\frac{a+j}{d}\right) = d \sum_{r=0}^n \binom{n}{r} (-\log d)^r \gamma_{n-r}(a) + \frac{(-1)^n d \log^{n+1} d}{n+1}. \quad (5.10)$$

The inhomogeneous term is the residue contribution, not an optional normalization.

6 Cyclotomic polylogarithm conductor descent

Put $\xi_d = e^{2\pi i/d}$. For $f \mid d$ and a primitive character χ^* of conductor f , define the primitive-root trace

$$P_{d,\chi^*}(s) = \sum_{a \in \mathcal{U}_d} \overline{\chi^*}(a) \operatorname{Li}_s(\xi_d^a). \quad (6.1)$$

At $d = 1$, the trace is $\zeta(s)$. At every $d > 1$, all roots in the sum are different from one, and the trace is entire in the order s . One proof of this last assertion is the finite Fourier formula

$$\operatorname{Li}_s(\xi_d^a) = d^{-s} \sum_{b=1}^d \xi_d^{ab} \zeta(s, b/d), \quad a \not\equiv 0 \pmod{d}. \quad (6.2)$$

The potential pole at $s = 1$ cancels because $\sum_b \xi_d^{ab} = 0$.

Let

$$B_{d,f,\chi^*}(s) = \sum_{r \mid R(d,f)} \mu(r) \overline{\chi^*}(r) \left(\frac{d}{fr}\right)^{1-s} = A_{d,f,\chi^*}(1-s). \quad (6.3)$$

The conjugation and the reflection $s \mapsto 1-s$ are both important.

Theorem 6.1 (Primitive-root trace descent). *As a meromorphic identity in s ,*

$$P_{d,\chi^*}(s) = B_{d,f,\chi^*}(s) P_{f,\chi^*}(s). \quad (6.4)$$

For $f > 1$ both sides are entire. For $f = 1 < d$, the right-hand side has a removable singularity at one and equals the entire left-hand side. For a primitive character,

$$P_{f,\chi^*}(s) = \tau(\overline{\chi^*}) L(s, \chi^*), \quad \tau(\overline{\chi^*}) = \sum_{a \in \mathcal{U}_f} \overline{\chi^*}(a) \xi_f^a. \quad (6.5)$$

Proof. It suffices first to work in $\operatorname{Re} s > 1$, where all Dirichlet series are absolutely convergent. Write $d = fm$. In the trace defining P_d , the restriction $(a, d) = 1$ can be imposed, after requiring $(a, f) = 1$, by inclusion–exclusion over $r \mid R(d, f)$. Put $a = rb$ in the term indexed by r . The inner exponential sum in the coefficient of n^{-s} is

$$\overline{\chi^*}(r) \sum_{b=1}^{d/r} \overline{\chi^*}(b) e^{2\pi i b n / (d/r)}.$$

Split b into classes modulo f . The sum vanishes unless $(m/r) \mid n$. When $n = (m/r)\ell$, it equals

$$\overline{\chi^*}(r) \frac{m}{r} \sum_{b \in \mathcal{U}_f} \overline{\chi^*}(b) e^{2\pi i b \ell / f}.$$

For a primitive character, the last sum is $\tau(\overline{\chi^*}) \chi^*(\ell)$, including zero when $(\ell, f) > 1$. Summing over n gives

$$P_{d,\chi^*}(s) = \tau(\overline{\chi^*}) L(s, \chi^*) \sum_{r \mid R(d,f)} \mu(r) \overline{\chi^*}(r) (m/r)^{1-s}.$$

The special case $d = f$ gives (6.5), and division is not needed to obtain (6.4). Analytic continuation and (6.2) finish the proof. The primitive Gauss identity used here is the standard finite Fourier identity for a primitive character; it follows by permuting units when $(\ell, f) = 1$, with primitivity forcing vanishing at nonunits. $\square$

Corollary 6.2 (All order derivatives away from the principal pole). *If $f > 1$, then for every $j \geq 0$ and every $s_0 \in \mathbb{C}$,*

$$P_{d,\chi^*}^{(j)}(s_0) = \sum_{r|R(d,f)} \mu(r) \overline{\chi^*}(r) m_r^{1-s_0} \cdot \sum_{h=0}^j \binom{j}{h} (-\log m_r)^h P_{f,\chi^*}^{(j-h)}(s_0). \quad (6.6)$$

For $f = 1$ the same formula holds away from $s_0 = 1$; at one, its finite part must be taken before evaluation.

Proof. Differentiate the finite exponential sum in (6.4). For $f > 1$ the factors are entire, so ordinary Leibniz differentiation applies at every order. $\square$

Example 6.3 (Two nontrivial level-twelve identities). For all complex s ,

$$P_{12,\chi_4}(s) = (1 + 3^{1-s})P_{4,\chi_4}(s), \quad (6.7)$$

$$P_{12,\chi_3}(s) = (4^{1-s} + 2^{1-s})P_{3,\chi_3}(s). \quad (6.8)$$

The first identity is

$$\begin{aligned} \text{Li}_s(\xi_{12}) + \text{Li}_s(\xi_{12}^5) - \text{Li}_s(\xi_{12}^7) - \text{Li}_s(\xi_{12}^{11}) \\ = (1 + 3^{1-s})(\text{Li}_s(i) - \text{Li}_s(-i)). \end{aligned}$$

These were checked directly at the nonintegral complex order $s = 1.7 + 0.3i$, separately from the proof.

7 Exact prime-support vanishing at order one

The principal trace is particularly transparent. Put

$$T_q(s) = \sum_{a \in \mathcal{U}_q} \text{Li}_s(\xi_q^a), \quad q > 1. \quad (7.1)$$

Theorem 6.1 gives

$$T_q(s) = q^{1-s} \prod_{p|q} (1 - p^{s-1}) \zeta(s). \quad (7.2)$$

This is also the Dirichlet series of the classical Ramanujan sum $c_q(n) = \sum_{a \in \mathcal{U}_q} \xi_q^{an}$, obtained directly by inclusion–exclusion. We use the product to examine the removable singularity, rather than substituting $s = 1$ into the factors separately.

Theorem 7.1 (Exact vanishing order). *Let $r = \omega(q)$ be the number of distinct prime divisors of $q > 1$. Then the entire function T_q has an exact zero of order $r - 1$ at $s = 1$. Here order zero means a nonzero value. Explicitly,*

$$\sum_{a \in \mathcal{U}_q} \mathcal{L}^{[j]}(\xi_q^a) = 0 \quad (0 \leq j < r - 1), \quad (7.3)$$

$$\sum_{a \in \mathcal{U}_q} \mathcal{L}^{[r-1]}(\xi_q^a) = (-1)^r (r - 1)! \prod_{p|q} \log p. \quad (7.4)$$

Proof. Set $s = 1 + \varepsilon$. Every factor $1 - p^\varepsilon = -\varepsilon \log p + O(\varepsilon^2)$ has a simple zero, while $\zeta(1 + \varepsilon) = \varepsilon^{-1} + O(1)$ has a simple pole. The nonzero factor $q^{-\varepsilon}$ equals one at zero. Therefore

$$T_q(1 + \varepsilon) = (-1)^r \left(\prod_{p|q} \log p \right) \varepsilon^{r-1} + O(\varepsilon^r). \quad (7.5)$$

The leading coefficient is nonzero because every prime exceeds one. Taylor coefficient extraction proves both statements. $\square$

7.1 Explicit identities

At level $q = 30$, $\mathcal{U}_{30} = \{1, 7, 11, 13, 17, 19, 23, 29\}$, and

$$\sum_{a \in \mathcal{U}_{30}} \text{Li}_1(\xi_{30}^a) = 0, \quad (7.6)$$

$$\sum_{a \in \mathcal{U}_{30}} \mathcal{L}^{[1]}(\xi_{30}^a) = 0, \quad (7.7)$$

$$\sum_{a \in \mathcal{U}_{30}} \mathcal{L}^{[2]}(\xi_{30}^a) = -2 \log 2 \log 3 \log 5. \quad (7.8)$$

At level 210, the first three traces, of derivative orders 0, 1, 2, vanish, and

$$\sum_{a \in \mathcal{U}_{210}} \mathcal{L}^{[3]}(\xi_{210}^a) = 6 \log 2 \log 3 \log 5 \log 7. \quad (7.9)$$

Prime powers do not change the first nonzero derivative: at $q = p^e$, $T_q(1) = -\log p$. Their exponents do affect later derivatives.

7.2 The next two coefficients, including repeated prime powers

Write

$$L = \log q, \quad S_1 = \sum_{p|q} \log p, \quad S_2 = \sum_{p|q} (\log p)^2, \quad C = (-1)^r \prod_{p|q} \log p,$$

and let $\gamma = \gamma_0(1)$, $\gamma_1 = \gamma_1(1)$. Expanding (7.2) two further orders gives

$$T_q^{(r)}(1) = r!C \left(\gamma - L + \frac{S_1}{2} \right), \quad (7.10)$$

$$T_q^{(r+1)}(1) = (r+1)!C \left[-\gamma_1 + \gamma \left(\frac{S_1}{2} - L \right) + \frac{1}{2} \left(\frac{S_1}{2} - L \right)^2 + \frac{S_2}{24} \right]. \quad (7.11)$$

To verify the expansion, use

$$\frac{e^x - 1}{x} = 1 + \frac{x}{2} + \frac{x^2}{6} + O(x^3), \quad \log \frac{e^x - 1}{x} = \frac{x}{2} + \frac{x^2}{24} + O(x^3),$$

and $\varepsilon \zeta(1 + \varepsilon) = 1 + \gamma \varepsilon - \gamma_1 \varepsilon^2 + O(\varepsilon^3)$. This gives an explicit all-prime-support family beyond the leading logarithmic identity. Higher coefficients are obtained by finite series multiplication with the same sign convention.

8 Twisted vanishing and a level-260 identity

Let χ^* be primitive of conductor $f > 1$, and $f \mid q$. Among the added primes, define

$$\mathcal{S} = \{p \mid q : p \nmid f, \chi^*(p) = 1\}, \quad r_\chi = |\mathcal{S}|.$$

The remaining added primes form $\mathcal{N}$.

Theorem 8.1 (Twisted leading jet). *The trace P_{q,χ^*} vanishes to order at least r_χ at one, and*

$$P_{q,\chi^*}^{(r_\chi)}(1) = (-1)^{r_\chi} r_\chi! P_{f,\chi^*}(1) \prod_{p \in \mathcal{S}} \log p \prod_{p \in \mathcal{N}} (1 - \overline{\chi^*}(p)). \quad (8.1)$$

Its order is exactly r_χ whenever $P_{f,\chi^}(1) \neq 0$. More generally its exact order is $r_\chi + \text{ord}_{s=1} P_{f,\chi^*}(s)$.*

Proof. The multiplier is

$$B_{q,f,\chi^*}(1 + \varepsilon) = (q/f)^{-\varepsilon} \prod_{\substack{p \mid q \\ p \nmid f}} (1 - \overline{\chi^*}(p) p^\varepsilon).$$

The factors indexed by $\mathcal{S}$ have simple zeros with leading coefficient $-\log p$. All other factors are nonzero at zero and have value $1 - \overline{\chi^*}(p)$. Multiply their leading terms by the Taylor series of $P_{f,\chi^*}(1 + \varepsilon)$. $\square$

For primitive nonprincipal characters, the classical nonvanishing $L(1, \chi^*) \neq 0$, together with the nonzero primitive Gauss sum in (6.5), supplies the stated nonvanishing hypothesis [7]. The explicit example below does not need that theorem; its primitive trace is computed directly from logarithms.

For $\chi_4(a) = (-1)^{(a-1)/2}$ on odd integers,

$$P_{4,\chi_4}(1) = \text{Li}_1(i) - \text{Li}_1(-i) = \frac{i\pi}{2}.$$

Take $q = 260 = 4 \cdot 5 \cdot 13$. Both added primes have $\chi_4(p) = 1$. Consequently

$$\sum_{a \in \mathcal{U}_{260}} \chi_4(a) \text{Li}_1(\xi_{260}^a) = 0, \quad (8.2)$$

$$\sum_{a \in \mathcal{U}_{260}} \chi_4(a) \mathcal{L}^{[1]}(\xi_{260}^a) = 0, \quad (8.3)$$

$$\sum_{a \in \mathcal{U}_{260}} \chi_4(a) \mathcal{L}^{[2]}(\xi_{260}^a) = i\pi \log 5 \log 13. \quad (8.4)$$

This is a proved identity, not an integer-relation conjecture. More generally, for distinct primes $p_1, \dots, p_r \equiv 1 \pmod{4}$ and $q = 4p_1 \cdots p_r$,

$$\sum_{a \in \mathcal{U}_q} \chi_4(a) \mathcal{L}^{[r]}(\xi_q^a) = (-1)^r r! \frac{i\pi}{2} \prod_{j=1}^r \log p_j, \quad (8.5)$$

with all lower order traces vanishing. The same leading identity persists if the added prime factors occur with higher exponents.

9 Evaluation without differentiating through a removable pole

The trace identities concern an analytic function at $s = 1$, but a numerical implementation of Li_s may internally use expressions whose gamma and zeta factors cancel only after substantial subtraction. Tiny-step black-box differentiation can then amplify roundoff.

There is an exact alternative. Expand (6.2) using (4.3) and remove the pole *symbolically* before extracting coefficients.

Proposition 9.1 (Pole-subtracted Fourier jet formula). *For $q \geq 2$, $a \not\equiv 0 \pmod{q}$ and $n \geq 0$,*

$$\mathcal{L}^{[n]}(\xi_q^a) = \frac{(-1)^n}{q} \sum_{b=1}^q \xi_q^{ab} \sum_{j=0}^n \binom{n}{j} (\log q)^{n-j} \gamma_j(b/q). \quad (9.1)$$

Proof. The pole coefficient is proportional to $\sum_b \xi_q^{ab} = 0$. The remaining product is

$$q^{-1} e^{-\varepsilon \log q} \sum_{b=1}^q \xi_q^{ab} \sum_{j \geq 0} \frac{(-1)^j}{j!} \gamma_j(b/q) \varepsilon^j.$$

Its n th Taylor coefficient, multiplied by $n!$, is (9.1). □

This formula is especially efficient for a whole primitive-root trace: precompute the Stieltjes table once and sum with its finite Fourier weights. For the principal trace those weights are the integer Ramanujan sums $c_q(b)$. It also avoids pretending that a floating-point sum of roots is *exactly* zero before multiplying by a numerical pole.

In the local audit, a direct tiny-step order differentiation of `mpmath.polylog` near order one produced a spurious approximately $1.4 \cdot 10^{-9}$ first trace at $q = 30$. The final evaluator did not loosen the acceptance tolerance. It instead used (9.1); all ten final smoke tests at 45 decimal working digits passed the 10^{-38} absolute-error threshold. The largest observed error was approximately $3.59 \cdot 10^{-43}$. This is an implementation observation for the tested version and settings, not a theorem about every differentiation routine.

An interval implementation remains desirable. The current numerical artifacts explicitly label themselves as floating-point smoke tests. The analytic identities are established by the proofs above, independently of those residuals.

10 An exact limitation of the tested laws for the S_4 candidate

The manuscript defines

$$S_4 = \sum_{n \geq 0} \frac{(-1)^n H_n}{(2n+1)^4}, \quad g_{a,b} = \text{Im Li}_{a,b}(i, 1),$$

and displays the candidate

$$S_4 = \frac{4g_{4,1} - 3g_{3,2} - 9g_{2,3}}{7} + \frac{\pi^5}{224} - \frac{27G\zeta(3)}{224} - 2\beta(4) \log 2, \quad (10.1)$$

where $G = \beta(2)$. We do not promote this identity to theorem status. We do, however, explain one precise obstacle to proving it from a restricted set of laws.

Lemma 10.1 (The colored double represented by S_p). *For $p > 1$,*

$$S_p = \text{Im} (\text{Li}_{p,1}(i, 1) + \text{Li}_{p,1}(i, -1)). \quad (10.2)$$

Proof. In the sum of the two doubles the inner color factor is $1 + (-1)^m$, which retains twice the even indices. At outer index $2n + 1$, the inner sum is $2 \sum_{j=1}^n (2j)^{-1} = H_n$, while $\text{Im}(i^{2n+1}) = (-1)^n$. Even outer indices contribute no imaginary part. Absolute convergence for $p > 1$ justifies the regrouping. $\square$

At weight five put

$$x_{a,b;c,d} = \text{Im Li}_{a,b}(i^c, i^d), \quad a + b = 5,$$

formally imposing conjugation $x_{a,b;-c,-d} = -x_{a,b;c,d}$. When both colors are real, the coordinate is zero. The six representative color pairs are

$$(0, 1), (1, 0), (1, 1), (1, 2), (1, 3), (2, 1).$$

There are thus 24 coordinates. In this paragraph all products of single polylogarithms and single weight-five values are quotiented out.

The tested presentation consists of the imaginary parts of the following two formal product rows, for $p + q = 5$ and $c, d \in \mathbb{Z}/4\mathbb{Z}$:

$$x_{p,q;c,d} + x_{q,p;d,c} = 0, \tag{10.3}$$

$$\sum_{j=0}^{p-1} \binom{q-1+j}{j} x_{q+j,p-j;d,c-d} + \sum_{j=0}^{q-1} \binom{p-1+j}{j} x_{p+j,q-j;c,d-c} = 0. \tag{10.4}$$

They are the depth-one-product stuffle and shuffle coefficient rows. The formal presentation includes the endpoint-color rows as well; restricting to convergent products alone can only reduce its row space. Nonzero rows, including duplicates, form a 96×24 integer matrix M . The ordinary level-two distribution sums of these color families have zero imaginary row after conjugation, so adding those rows does not alter this matrix rank.

By Lemma 10.1, the depth-two left side of seven times (10.1) is

$$\ell = 7x_{4,1;1,2} + 3x_{4,1;1,0} + 3x_{3,2;1,0} + 9x_{2,3;1,0}. \tag{10.5}$$

Its conjectured single-value right side is $\pi^5/32 - 27G\zeta(3)/32 - 14\beta(4)\log 2$.

Proposition 10.2 (A finite-presentation obstruction). *The row ℓ is not in the row space of the presentation (10.3). In fact,*

$$\text{rank } M = 20, \quad \text{rank} \begin{pmatrix} M \\ \ell \end{pmatrix} = 21.$$

Thus (10.1) cannot follow merely by linear combination of the tested same-weight single-product rows, after the specified single-value quotient. This is not a disproof of the candidate.

Proof. It suffices for the nonmembership assertion to give a column vector v annihilated by all rows of M but not by ℓ . Its only nonzero entries are the following:

a	b	c	d	$v_{a,b;c,d}$
1	4	1	0	-2
2	3	0	1	1
2	3	1	0	3
2	3	1	3	-1
3	2	0	1	-3
3	2	1	0	-1
3	2	1	3	-1
4	1	0	1	2

Substitution in the finite binomial sums (10.3), with the stated conjugation rule, gives $Mv = 0$. On the other hand, $\ell v = 3(-1) + 9(3) = 24$. Hence ℓ is not a row combination. The two displayed ranks are separately verified by exact integer/rational row reduction in `verify_s4_obstruction.py`. The full matrix, row labels and witness are retained in the JSON certificate; no numerical values of polylogarithms are used in this check. $\square$

10.1 What remains plausible and what was numerically checked

The conjecture was independently tested using the one-dimensional integrals

$$S_4 = -\frac{1}{6} \int_0^1 \frac{(-\log t)^3 \log(1+t^2)}{1+t^2} dt, \quad (10.6)$$

$$g_{a,b} = \frac{1}{(a-1)!} \operatorname{Im} \int_0^1 (-\log t)^{a-1} \frac{i \operatorname{Li}_b(it)}{1-it} dt. \quad (10.7)$$

The first follows from the harmonic-number generating function; the second from the derivative of $\operatorname{Li}_{1,b}(z, 1)$ and repeated logarithmic integration. An initial 50-digit check gave residual magnitude about $2.32 \cdot 10^{-51}$; the packaged 45-digit rerun also passed. This is evidence, not a proof.

The exact obstruction says where the next proof attempt must look: an additional functional equation, a higher-depth relation whose elimination creates a new double relation, or an argument transformation outside the single-product presentation. Enlarging the coefficient height in the same law span cannot solve the problem. It would be incorrect either to reject (10.1) because of this obstruction or to claim it is proved because its floating-point residual is small.

11 Corrections and integration recommendations

11.1 Replace the finite rank observation by a precisely scoped theorem

The most substantial editorial change is to replace `tower:thm:rank` with Corollary 4.2, including Definition 2.1. State that the theorem concerns the complete prime/distribution presentation with the endpoint fixed. Do not replace it by a bare numerical-independence assertion. The standalone replacement fragment is supplied in the integration directory.

11.2 Make the weight-one endpoint explicit

Some prose surrounding the Gaussian and Eisenstein single values says that their real parts are rational multiples of $\zeta(n)$. The $n = 1$ case must be separated or interpreted by a removable limit. In the principal branch,

$$\operatorname{Li}_1(i) = -\frac{1}{2} \log 2 + \frac{i\pi}{4}, \quad (11.1)$$

$$\operatorname{Li}_1(e^{2\pi i/3}) = -\frac{1}{2} \log 3 + \frac{i\pi}{6}. \quad (11.2)$$

For example, $\frac{1}{2}(3^{1-s} - 1)\zeta(s)$ tends to $-\frac{1}{2} \log 3$ at $s = 1$; literal substitution gives an undefined zero times pole. This is a domain clarification, not evidence that the higher-weight formulas are wrong.

11.3 Do not invert a vanishing conductor multiplier

A formula that solves for Z_{f,χ^*} by dividing Z_{q,χ^*} by $A_{q,f,\chi^*}(s)$ needs a nonzero-denominator hypothesis. The polynomial normal form avoids it. The same warning applies to the polylogarithm multiplier B_{q,f,χ^*} , where the zeros at one produce the identities in Sections 8 and 9. A disappearing primitive-root trace is not a disappearing formal character direction.

11.4 Keep the undifferentiated pole term

Any extension of the manuscript's $k \geq 1$ identities to $k = 0$ must include Proposition 5.2 for the principal character. The extra term is forced by the Laurent pole. Its omission would already fail at $n = 0$.

11.5 Preserve the status of the remaining S_4 candidate

The earlier weight-five sporadic relation and weight-six Gaussian parity reductions are not represented here as new work. The displayed S_4 formula remains a candidate in this contribution. Add its exact single-product obstruction as a methodological note, not as a negative numerical claim. A later proof using a larger law vocabulary would be fully compatible with the certificate given here.

12 Reproducibility, verification and implementation boundaries

The package contains three kinds of evidence. First, this article proves the general statements without relying on finite experiments. Second, exact scripts verify finite matrices and a polynomial normal form. Third, independent high-precision numerical representations test signs, conjugations and coefficient conventions.

The completed exact run contains 295 specialized rank cases: $2 \leq q \leq 60$ at weights $t_p = 1, p^2, p^3$, at all-zero prime weights, and at one mixed integer assignment extended multiplicatively. Each checks both the full matrix and its endpoint-fixed projection. There are 29 all-divisor checks for $2 \leq q \leq 30$, and 18 symbolic jet cases at $q \in \{4, 6, 8, 9, 10, 12\}$ and orders 1, 2, 3. The jet logarithms are independent symbols in these finite checks; the mathematical theorem establishes validity after arbitrary specialization.

The level-twelve polynomial check verifies both identities in (3.14). The S_4 check constructs all 96 rows from the two binomial rules, proves the witness equation, and computes the two ranks. The exact run also records the first nonzero principal trace jets for six levels. A separate finite Taylor verifier checks the next two principal coefficients at those levels and the twisted level-260 leading coefficient. JSON outputs record dimensions, coefficients and software versions.

The ten numerical checks use 45 decimal working digits. They include the three level-30 traces, two twisted level-20 traces, two complex-order conductor identities, two Stieltjes descent identities, and the retained S_4 candidate. They are neither interval enclosures nor proofs of equality. The explicit level-260 identity is proved analytically; this package does not claim to have performed a separate 96-term high-precision evaluation of that example.

Artifact	Purpose
<code>src/distribution.py</code>	Explicit matrices, symbolic jet rows, level-twelve normal form.
<code>src/verify_exact.py</code>	Exact rank and polynomial checks, with JSON output.
<code>src/verify_numeric.py</code>	Pole-subtracted floating-point smoke tests.
<code>src/verify_trace_coefficients.py</code>	Finite Taylor checks for principal and twisted traces.
<code>src/verify_s4_obstruction.py</code>	Independent finite law-span obstruction.
<code>integration/</code>	Replacement theorem and editorial notes, not an automatic repository write.
<code>MANIFEST.sha256</code>	Integrity hashes for the distributed package.

No remote repository file was modified. The integration fragments require normal editorial review, particularly comparison with any historical matrices recovered later. The Python generator uses exact rational or symbolic domain arithmetic for rank; there is no numerical rank tolerance. It is a reference implementation, not a bit-complexity claim or a production-scale character-table engine.

13 Further research questions

13.1 An integral version of the polynomial normal form

The proof uses character idempotents and division by group orders. Over a splitting field this is harmless; over $\mathbb{Z}[t_p]$ it hides torsion and integral-lattice questions. Determine an integral presentation adapted to prime powers, and characterize the Smith data after integer-weight specialization. Classical ordinary-distribution results suggest useful structures, but the present field-level proof does not establish integral freeness for this deformation.

13.2 The conductor transition determinant

The matrix from minimal-conductor generators to level- q character sums is diagonal with entries A_{q,f,χ^*} . Its determinant and its zero divisor are therefore explicit. A systematic treatment should relate the vanishing multiplicities to arithmetic splitting of added primes, and should distinguish coordinate singularities from intrinsic period relations. This is a good place to develop exact conditioning estimates for character-based numerical algorithms.

13.3 Certified order jets at roots of unity

Implement Proposition 9.1 with ball arithmetic for Stieltjes constants and exact cyclotomic weights. Derive a precision budget that accounts for cancellation in a trace with a prescribed high-order zero. The finite Fourier formula supplies an exact target; the missing part is a sharp certified numerical analysis and a tested interval implementation.

13.4 A proof of the S_4 identity outside the tested row span

The witness in Proposition 10.2 excludes one route, not the identity. A focused next step is to add weight-five higher-depth relations and eliminate the triple coordinates, or to seek a

functional transformation of the integrals (10.6). A successful proof should provide an exact law certificate whose new row does not annihilate the witness. The candidate must remain explicitly conjectural until then.

13.5 Higher-depth conductor descent

At depth one, removing an added prime is controlled by an Euler factor. For colored double and triple sums, prime divisibility interacts with the strict inequalities between indices. Determine a finite operator calculus that records these interactions and produces a conductor-adapted filtration. A straightforward product of the depth-one factors should not be assumed to work; the collision terms are the mathematical content of the problem.

13.6 Formalization of the rank theorem

A proof-assistant development can be divided into finite character orthogonality, the two prime-edge identities, a commuting divisor-lattice normal form, and base change to truncated germs. The rank theorem itself requires no transcendental arithmetic. The analytic realization can be formalized later, using the established Hurwitz multiplication and parameter-derivative laws.

13.7 Numerical independence remains separate

The conductor quotient supplies exact formal coordinates, not a lower bound for the dimension of their evaluated periods. A meaningful independence conjecture must state the coefficient field, which lower periods are quotiented out, and which order derivatives are included. Special logarithmic identities and functional equations must first be removed. Failed integer-relation searches do not fill this gap.

14 Conclusion

The unresolved distribution-rank pattern can be proved uniformly once the presentation is stated and the denominator lattice is organized by minimal character conductors. The resulting module is polynomially free, so neither specialization nor passage to arbitrary jets creates a rank jump. This gives an explicit theorem for the manuscript's complete distribution system and an all-order descent formula for its Stieltjes tower.

The same structure, reflected by $s \mapsto 1 - s$ and character conjugation, produces exact cyclotomic polylogarithm trace identities. At order one, prime support controls the first nonzero derivative; the residue of the Riemann zeta function lowers the principal vanishing order by exactly one. These identities also expose why order-one numerical evaluation must remove poles symbolically.

Finally, the S_4 certificate separates a plausible identity from an insufficient collection of proof laws. Together, the positive normal-form theorems and this finite obstruction give concrete next steps without confusing formal rank, numerical evidence and arithmetic independence.

A Exact normal-form algorithm in conductor coordinates

A symbolic implementation need not perform Gaussian elimination on all q residue coordinates. The following procedure describes the mathematics of Theorem 3.1.

For each divisor $d \mid q$, compute the character transform on $\mathcal{U}_d$. Attach to each transformed coordinate its primitive conductor f . Replace that coordinate by $A_{d,f,\chi^*}(\mathbf{t})Z_{f,\chi^*}$ using (3.2). Collect identical primitive-character generators. For order jets, replace t_p by its truncated exponential and multiply polynomials modulo ε^{N+1} . Finally invert the finite character transforms only for the original residue coordinates actually requested.

Every operation is finite and every transition multiplier is polynomial. The coefficient field may be chosen as a cyclotomic field containing the character values. Complex characters must be paired or represented exactly; rounding them to floating-point roots of unity defeats the exactness of the method. The reference package implements the full residue matrices and the entire level-twelve normal form, while leaving a general optimized character-table implementation as a subsequent engineering task.

B Source snapshot and claim ledger

The repository branch was resolved during this audit to commit

85c89e5d0bf785c058bd40fba286588e32a11382.

The inspected Chapter 8 blob is

0a5b5ec8b7cb2213302c5a0b857a43e0f99b6236.

The audit targets are the rank observation in Chapter 8, the surviving research programme in Chapter 10, and the S_4 display in Chapter 4. The source’s earlier provenance and its independently prepared research packages are not overwritten by this note.

Claim	Status in this package
Multiparameter conductor freeness	Proved, Theorem 3.1.
Uniform endpoint-fixed rank for the complete presentation	Proved, Corollaries 3.3 and 4.2.
Equality with an unavailable historical script’s exact row list	Not asserted.
All-order jet dimensions	Proved, Theorem 4.1.
All-order Stieltjes conductor identities	Proved, Theorem 5.1; $k = 0$ correction explicit.
Principal and twisted trace identities	Proved, Sections 7–9.
Global novelty of universal-distribution rank phenomena	Not asserted; classical literature credited.
S_4 displayed closed form	Numerically supported, not proved here.
Obstruction in the specified 96×24 presentation	Exactly certified; not a disproof of S_4 .
Independence of evaluated periods	Not asserted.
Formal verification in Lean or another proof assistant	Not performed.
Rigorous numerical interval certification	Not performed.
Remote repository integration	Not performed; fragments supplied.

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